

Spoken Tutorial on First Project

Module	An introduction to Building
Episode	3. First Project
Learning Objective	At the end of this tutorial learner will be able to 1. Define the purpose of a project setup. 2. Identify steps to set up a development environment. 3. Apply the Gemini API key for authentication. 4. Explain how to create and send an API request.
Approx. Duration	3-4 min
Outline	• Project Overview • Obtaining an API Key • Setting Up Google Colab and Dependencies • Configuring the Gemini API and Model • Crafting and Sending an API Request • Displaying AI Responses
Meta Tags	Gemini API, Google Colab, AI Project, API Key, Prompt Engineering, Spoken Tutorial, AI Model, API Request, Python, Development Setup, Generative AI
Pre-requisite Tutorial	Episode 1 and 2 of this Module

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on First Project .
Learning Objectives Slide	In this tutorial, you will learn to, • Define project setup. • Set up a development environment. • Apply the Gemini API key. • Create and send an API request.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have completed Episode 1 . Also, you should have completed Episode 2 of this Module .
Illustration showing project overview with blueprint and house icons	What is a project overview ? Think of it like a blueprint for building a house. It outlines what you will build and why. It also shows the main components involved. This ensures everyone understands the goal. This happens before you even start.

Browser showing Google AI Studio interface with 'Get API Key' section highlighted	First, navigate to the Google AI Studio . Locate the 'Get API Key ' section there. Then, generate a new API key . This key grants your project access to Gemini's features. It is essential for authentication.
Screen close-up: user typing 'pip install google-generativeai...' in Google Colab	Next, open Google Colab in your browser. Install necessary Python packages. For example, use `pip install google-generativeai`. This sets up your development environment.
Screen close-up: user typing 'import google.generativeai...' and configuring API key	Import the `google.generativeai` library. Configure it with your API key . Then, initialize the Gemini AI model you want to use. This prepares the model for interaction.
Screen close-up: user typing 'write a product description' into a prompt input field	Now, let's craft your first prompt . Define a product topic, like 'make a new product'. Then, create a simple, unspecific prompt . For example, 'write a product description'.
Screen close-up: user typing 'Write a compelling product description for a smart reusable coffee cup...' into a prompt input field	Let's improve this prompt . Refine your product topic to be specific. For instance, choose 'a smart reusable coffee cup'. Now, create a detailed prompt . It should ask for a compelling product description. Remember to highlight its eco-friendliness. Also, include temperature control and app integration.
Side-by-side: vague prompt output vs improved prompt output	Compare the two prompts . Notice how the improved prompt yields a richer description. It is also more relevant. The vague one gives a general response.
AI generating response on screen, showing the output of the improved prompt	Use the initialized model to send your prompt . This goes as an API request . Once the request is processed, display it. See the generated AI response .
Summary Slide	Let us summarize what we learned. We set up a development environment. We configured the Gemini API . Then, we crafted effective prompts . Finally, we retrieved AI responses .
Assignment Slide	Now, for an assignment, try creating a new product topic. Develop a more detailed prompt for it. Generate a unique description. Experiment with different model parameters. Compare the results.

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